MILITARY SPECIFICATION

HOWITZER, MEDIUM, TOWED, 155MM: Ml 98

This specification is approved for use by U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 scope. This specification covers one type of 155MM, medium, towed howitzer, M198: consisting of the M199 cannon, M45 recoil mechanism, M39 carriage and associated fire control.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following documents, of the *issue* in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-14232	-Parts, Equipment and Tools for Army Material, Packaging and Packing of
MIL-I-45607	-Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-45975	-Cannon, 155MM Howitzer: M199
MIL-M-45986	-Mechanism, Recoil, 155MM Howitzer: M45
MIL-C-45987	-Carriage, 155MM Howitzer: M39
MIL-A-48449	-Alignment Device, Optical, Gun Bore
MIL-T-48554	-Telescope, Panoramic: M137
MIL-Q-48555	‘Quadrant, Fire Control: M17
MIL-Q-48556	-Quadrant, Fire Control: M18
MIL-M-48557	-Mount, Telescope and Quadrant: M171
MIL-T-48558	-Telescope, Elbow: M138
MIL-M-48559	-Mount, Telescope and Quadrant: M172
MIL-W-63150	-Weapons and Support Materiel Standard Quality Assurance Provisions for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center,Attn. DRSMC-OA, Dover,New ersey07801 by using the self-addressed Standardization Document Improvement Proposal (DDForm 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED.- STD. NO. 595	-Colors

MILITARY

MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-130	-Identification Marking of U.S. Military Property
MIL-STD-171	-Finishing of Metal and Wood Surfaces
MIL-STD-1458	-Radioactive Materials: Marking and Labeling of Items, Packages and Shipping Containers for Identification in Use, Storage and Transportation

2.1.2 Other Government documents, drawings and publications.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

K12008000	-Howitzer, Medium, Towed, 155MM: M198

(Product drawings referenced in this specification form a part of Drawing K12008000).

PUBLICATIONS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

P12008000	-Packaging Data Sheet for Howitzer, Medium, Towed, 155MM: M198 W/E
LQ12008000	-Quality Requirements List, Howitzer, Medium, Towed, 155MM: M198
EL12008000-I	-List of Inspection Equipment Numbers, Howitzer, Medium, Towed, 155MM: M198

U.S. ARMY

AR 702.3	-Army Materiel Reliability, Availability and Maintainability (RAM)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.1.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Y14.5 -Dimensioning and Tolerancing

(Applications for copies should be addressed to· The American Society of Mechanical Engineers, united Engineering Center, 345 East 47th Street, New York, NY 10017).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be specified in the contract (see 6.2). Unless otherwise specified (see 6.2), the first article shall include the pilot pack (see 5.1).

3.2 Materials, construction and design. The howitzer shall conform to the materials, construction and design requirements specified herein and on Drawing K12008000.

3.2.1 Dimensions. Dimensions and tolerantes of parts and assemblies of the howitzer shall conform to the detailed drawings of Drawing K12008000. Interpretation of the dimensions and tolerances shall be in accordance with ANSI Y14.5.

3.2.2 Protective finishes.

3.2.2.1 Inorganic finishes. Parts requiring an inorganic finish shall be finished or coated as indicated on the applicable drawings and in accordance with MIL-STD-171.

3.2.2.2 Painting The parts and assemblies shall be painted as specified on the drawing and in accordance with MIL-STD-171. The visible surfaces of attaching hardware (nuts/screws, washers, etc.), exposed unpainted surfaces Of mounting faces, or blemished areas shall be primed and painted to match surrounding surfaces after final assembly. To avoid necessity of painting exposed surfaces of cadmium plated hardware (including screws, nuts and washers), the hardware may be dyed with a supplementary chromate finish, Olive Drab 24087 of Fed. Std. No. 595.

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specified on the applicable. threaded fasterners shall be torqued in accordance with the following torque requirements:

Thread Size	Torque
No. 4-48UNF	6-8 in-lb
No. 6-32UNC	7-9 in-lb
No. 8-32UNC	14-17 in-lb
No. 10-24UNC	20-25 in-lb
No. 10-32UNF'	20-25 in-lb
.250 - 20UNC	70-80 in-lb
.375 - 16UNC	20-24 ft-lb
.500 - 13UNC	50-60 ft-lb
.625 - llUNC	90-119 ft-lb

3.3 Performance characteristics.

3.3.1 Functioning. The howitzer, with a dummy round of ammunition (approximate weight 103 pounds) inserted in the chamber, shall meet the requirements specified in 3.3.1.1 through 3.3.1.7 when placed on a level surface. Testing shall be as specified in 4.5.3.2.

3.3.1.1 Elevating mechanism. The elevation mechanism shall elevate the cannon tube of the howitzer from a position of minus 75 roils depression to plus 1,275 roils in elevation. The torque required to elevate or depress at any point throughout the operating range shall not exceed 180 inch-pounds at the handwheel shaft. The maximum total system backlash of the elevating handwheel shall not exceed 30 degrees of rotation a any position in either direction of elevation or depression. Total system backlash is defined as the total handwheel lost motion until first movement of *he cradle, when the elevating handwheels are utilized to elevate and depress the weapon.

3.3.1.2 Traversining mechanism.

3.3.1.2.1 With the howitzer placed in the firing positio and the trails in the open position, the traversing mechanism shall be capable of traversing the top carriage 22-1/2 degree 1/2 degree left and right of center. The traverse stops shall not be removed during this test. The torque required to traverse the top carriage left and right of center shall not exceed 120 inch-pounds on the handwheel shaft.

shall not exceed 30 degrees of retation in theer direction of the handwheel while traversing a maximum of 2-1/2 degree + 1/2 degree left and right of center.

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3.3.1.2.3 With the trails in the closed position, the traversing mechanism shall be capable of traversing the top carriage 360 degrees in either direction.

3.3.1.3 Speed shift assembly. With the howitzer placed in firing position with trails in open position, the speed shift assembly shall be capable of raising the howitzer to permit manual rotation of the carriage 360 degrees. The maximum force required on the hydraulic pump handle to actuate the speed shift shall not exceed 12 pounds. The speed shift assembly shall be capable of maintaining a controlled rate of descent to the firing base of not less than 5 seconds and not more than 20 seconds during gravity drop. Sampling and testing shall be in accordance with paragraphs 4.5.3.7 and 4.5.7.3, respectively.

3.3.1.4 Variable recoil mechanism. The variable recoil mechanism shall be assembled and adjusted in accordance with Drawing K12007600 and K1200BOO0. When elevating or depressing the cannon, the mechanism shall move freely without binding or looseness. Adjustment of the linkage shall be as fooows:

3.3.1.4.l. Short recoil. The 55 degree (from vertical) mark on end of recuperator cylinder shall lie within the notch on end of regulator lever when the cannon is at 1025 mils elevation or more.

3.3.1.4.2 Long recoil. The 05 degrees (from vertical) mark on end of recupoerator cylinder shall lie within the notch on end of regulator lever when the cannon is at minus 50 mils elevation or less

3.7.1.5 Wheel actuator. The hydraulic wheel actuator shall be capable of lowering the carriage to the firing position and raising the carriage to the travel position. The maximum force required on the hydraulic pump handle to raise and lower the carriage shall not exceed 32 Pounds. During gravity drop of the carriage, a controlled rate of descent to the firing base shall not be less than 5 seconds nor qreater than 20 seconds

3.3.1.6 Locking devices. All locking and fastening devices for securing component parts shall be positive in action.

3.3.1.7 Handbrakes. The handbrakes, when engaged, shall prevent wheel rotation while the howitzer is in an ascending or descending travel position on a 10 degree slope having a surface on which the tires shall not slip. Travel position shall be defined as with the trails locked in closed position and the lunette raised 30 inches above ground level. When the handbrake is disengaged, the wheel-s shall turn freely with minimum drag.

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3.3.2 Fire control.

3.3.2.1 Gun elevation -plumb line level. The elevating trunnions of the carriage shall be cross leveled by having the cannon track a plumb line within 0.1 mil through a range of O to 978 roils elevation minimum. The fire control instruments shall meet the requirements of 3.3.2.1.1 through 3.3.2.6. Testing shall be as specified in 4.5.3.3.

3.3.2.1.1 All knobs on the fire control instruments shall function properly without binding, looseness or excessive torque.All counters shall operate properly and all radioactive sources shall be illuminated, including those on the off-carriage equipment.

3.3.2.2 Alignment of mount M171, panoramic telescope M137 and guadrant

a. The quadrant elevation level vial shall be adjusted, as necessary, to center the bubble within the thickneww (width) of a graduation line when the cannon tube is at zero elevation and the elevation counter is set at 0000.

b. After boresighting on the target and setting the azimuth counter to 3200 roils, the panoramic telescope shall align on the M139 alignment device at 4800 roils within + 0.5 roil.

c. Elevate the cannon tube from O to 1275 roils in 200 mil steps. Relevel the M171 in pitch and cross-level to within the thickness (width) of a graduation line for each 200 mil step. The vertical reticle line of the telescope shall not deviate from the test target vertical line by more than + 0.6 mil from O to 899 roils elevation and not more than + 1.5-mils from 900 to 1275 roils elevation.

3.3.2.3 Alignment of mount M172 and quadrant M18.

a. The elevation pads on the mount shall be parallel with the cannon tube within + 0.2 mil.

b. The quadrant elevation level vial shall be adjusted, as necessary, to center the bubble to within the thickness (width) Of one graduation line when the cannon tube is at zero elevation and the elevation counter is set at 0000.

3.3.2.4 Accuracy of elevation counters. Prior to proof firing, the elevation counter of the M17 and M18 quadrants shall reflect tube elevation and shall not exceed 1.0 mil deviation, excluding backlash, at any counter setting. After proof firing, the allowable deviation shall not exceed 1.5 mils, excluding backlash. The M18 cross level vial bubble shall remain centered within the thickness (width) of a graduation line through all tube elevations.

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3.3.2.5 Backlash. Backlash shall not exceed 1.0 mil at any elevation counter reading.

3.3.2.6 Illumination. The reticle markings, counter numerals, and level vial bubble and graduation lines shall be clearly distinguishable when observed in ambient light conditions ranging from dusk into darkness.

3.3.3 Proof firing. The howitzer shall withstand proof firing as specified in 4.6.3 with no evidence of system malfunction, structural damage, permanent deformation or degradation of operating characteristics. The recoil system shall meet the following criteria:

3.3.3.1 Long recoil length. Maximum long recoil length shall not exceed 72 Inches.

3.3.3.2 Short recoil length. Maximum short recoil length shall not exceed 52 inches.

3.3.3.3 Rod pull. Maximum rod pull (total of both recoil rods) shall not exceed 83,000 pounds.

3.3.3.4 Cycle time. Recoil cycle time shall not be less than 2.0 seconds and not more than 4.0 seconds except for 700 roils elevation and greater (high elevation firing); it shall not exceed 6.0 seconds.

3.3.3.5 Rod pull curves. The actual rod pull curves shall meet the following requirements: (see Figure 1)

Range 1: Peak rod pull shall occur within the first 20 miliseconds for long recoil and 60 milliseconds for short recoil.

Range 2: Within 280 milliseconds (max-) for long recoil or 180 milliseconds (max.) for short recoil, the continuous rod pull curve shall not exceed the peak rod pull of Range 1.

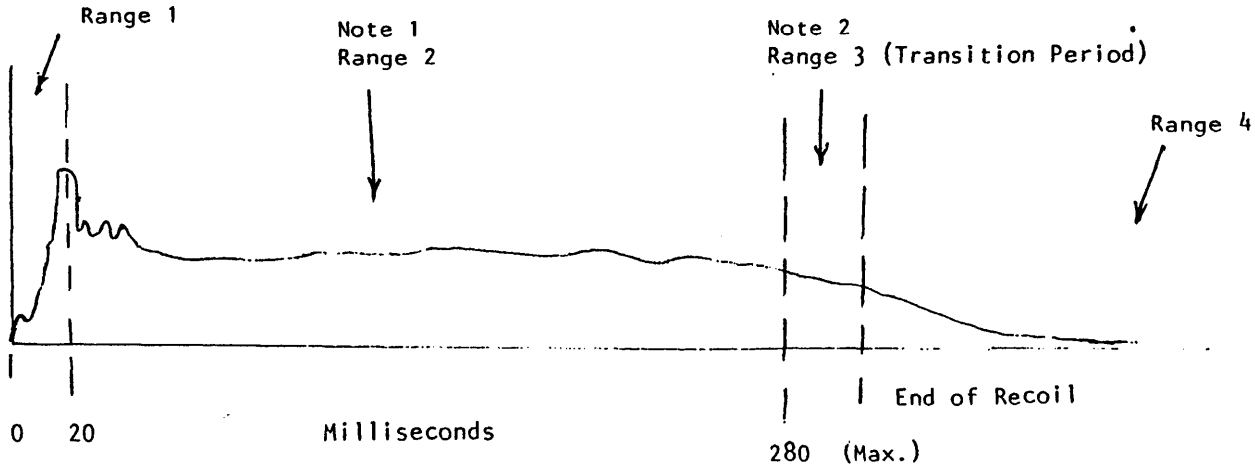
Range 3: After Range 2, the rod pull curve shall be continuously declining to the end of recoil cycle.

Range 4: Counter recoil cycle. The terminal velocity shall not exceed 15 inches/second on the last 6 inches of travel.

3.3.4 Road test. The howitzer shall be caple of withstanding the following 50 mile road test without damage permanent deformation or loosening of parts. Testing shall be as specified in 4.5.3.5.

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LONG RECOIL



Note 1: Rod pull trace is allowed to increase or decrease within the range as long as it does not exceed the peak in Range 1.

Note 2: Rod pull trace must be continuously decreasing in this range to the end of the recoil cycle.

SHORT RECOIL

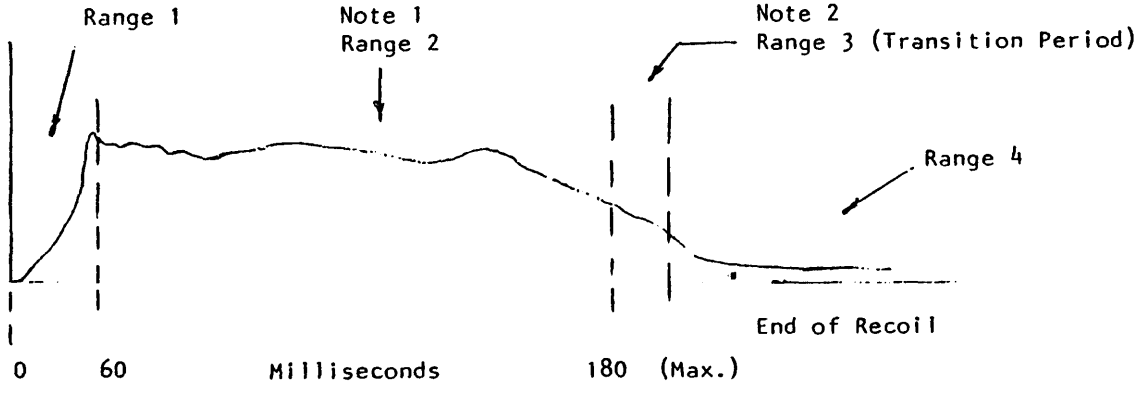


FIGURE 1 Typical rod pull curves

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-25 miles at speeds between 10 and 55 mph over hard surface or improved roads.

-20 miles at speeds between 10 and 15 mph over Belgian block or equivalent.

-5 miles at speeds between 10 and 15 mph over cross country roads.

3.3.4.1 Brake system. The brake system, when attached to the prime mover, shall provide efffective and controlled braking of the howitzer when stopping or slowing down from speeds up to 55-mph on improved roads and when descending slopes up to 60 percent. The break away emergency actin of the brakes shall be capable of locking the wheels when tehe howitzer and air hosesl break away from the prime mover.

3.3.5 Boresight retention. After proof firing and road tests, the line of sight of the M137 panoramic telescope shall maintain its original setting in azimuth with respect t the cannon tube within 0.5 mil. The M138 elbow telescope shall maintain its orginal setting with respect to the cannon tube within 0.8 mil in azimuth and 0.8 mil in elevtion. Testing shall be asspecified in 4.5.3.6.

3.4 Reliability, maintainability and durability. Howitzers manufactured in accordance with this specification shall be free from any defects that compromise, limit or reduce the howitzer capability specified herein and shall be subjet to examination and test by the Government to prove howitzer capaibities and performance in accordane with following:

3.4.1 Reliability. The reliability requirement is a specified value of 1100 mean rounds betwen failure and a minimum acceptable value of 700 mean rounds between failure as defined in AR 702.3.

3.4.2 Maintainability. The maintainability requirement is 0.5 hours for mean times to repair (MTTR) at organizational and 2.0 hours for direct support.

3.4.3 Durability. The durability requirement is 15,000 Equivalent Full Charge (EFC) rounds without rebuild or overhaul.

3.5 Howitzer components. The M45 recoil mechanism shall conform to MIL-M-45986, the M39 carriage shall conform to MIL-C-45987, the M189 cannon shall conform to MIL-C-45975, the M139 alignment device shall conform to MIL-A-48449, the M137 panoramic telescope shall conform to MIL-T-48554, the M17 quadrant shall confrom to MIL-C-48554, the M17 conform to MIL-Q-48556, the M171 mount shall conform to MIL-M-48557, the M138 elbow telescope shall conform to MIL-T-48558 and the M172 mount shall conform to MIL-M-48559.

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3.6 Identification and insturctional marking. Identification and instructional markings shall be applied in accordance with Drawing K120 08000. Other marking, including application of part numbers and instructional marking, shall be as specified on the applicable drawings and in accordance with MIL-STD-13. Serial number shall be furnished by the procuring activity (see 6.2).

3.7 Regulations pertaining to radioactive devices. Marking, Identification, handling and shipping of radioactive devices shall be in accordance with MIL-STD-1458.

3.8 Workmanship. Workmanship and finish shall be in accordance with the highest grade practice used in manufacturing military weapons. Finished items and parts shall not exhibit poor material and processing such as seams, laps, laminations, cracks, visible steps, sharp edges, nicks, stratches, burrs. deformations and missing operations which may affect serviceability, functioning, operation, appearance or Safety. Fins and other extraneous metal shall be removed from cast or forged parts. Hammering to shape, salvage operations (including repair by welding) or other similar practices shall not be permitted without prior approval of the contracting officer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The contractor shall perform, as a minimum, inspections in accordance with the specifications, quality assurance provisions, MIL-W-63150 and such documents as specified in LQ12008000, and the contract. These minimum inspections shall not be construed as relieving the contractor of his responsibilities under terms of the contract to furnish the Government with items complying with and conforming to the requirements of the contract, drawings and specifications.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

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4.3 Classification of inspection. The inspection requirements specified herein are as follows:

First article inspection (see 4.41.
b. Quality conformance inspection {see 4.5).

4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with the contract (see 6.2). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein and such other inspection as necessary to determine that all requirements of the contract have been complied with. One of the first article items will be subjected to a reliability, maintainability and durability demonstration (see 4.7), performed by the Government for informational purposes.

4.5 Quality conformance inspection.

4.5.1. Sampling. Each howitzer shall be inspected for compliance with the specification except sampling for speed shift assembly test, proof-firing test, road test, and boresight retention test shall be:

The first three production howitzers.
One howitzer per lot after successful completion of paragraph 4.5.1.a

A howitzer lot shall consist of two months production or thirty-five howitzers, whichever quantity is -smaller.

4.5.2 Examination.

4.5.2.1 Howitzers. Visually and manually examine each howitzer to determine conformance with the applicable requirements. Each step in the examination shall include a visual examination for proper cleaning, the presence of the specified protective coating, and to determine the general quality, completeness of manufacture, assembly and workmanship (see 3.8). Visually examine all markings to assure that they are correct and legible (see 3.6).

4.5.2.2 Parts and assemblies. Parts and assemblies shall be visually and manually examined to determine compliance with the materials, construction and design requirements of 3.2 and the applicable requirements of 3.2.l. through 3.2.3. Visually examine for marking (see 3.6) and workmanship (See 3.8). Parts and assemblies failing to comply with requirements shall be rejected.

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4.5.2.2.1. Protective finishes. Examination of painted surfaces and inorganic finishes shall be in accordance with the specifications specified on the applicable drawings and MIL-STD-171.

4.5.2.2.? Marking. Visually examine the identification plate for compliance with the applicable drawing, for secure assembly, serialization and to assure that all pertinent data has been applied.

4.5.3 Testing.

4.5.3.7 Failire data. Unless otherwise specified herein, all tests shall be conducted on a complete howitzer. If test requirements cited herein are not complied with, acceptance of the howitzer or howitzer lot shall be deferred and the contractor shall accomplish, as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional, physical and visual examination of the components which are suspected to be the cause of failure or malfunction.

b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.

c. Examine howitzers, partially assembled howitzers, and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.

d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned howitzers for retest.

4.5.3.2 Functioning testing. Each completely assembled howitzer" shall be tested for functioning (see 3.3.1 and 3.3.1.1 through 3.3.1.8) using the test methods-specified in 4.6.1.1, 4.6.l.2, 4.6.1..4, 4.6.1.5, 4.6.1.6, 4.6.1.7 and 4.6.7.8. Failure of the howitzer to comply with the aforementioned requirements shall be cause for rejection. Additionally, one completely assembled howitzer shall be tested per 4.5.1 as specified in 4.6.1.3. Failure of the howitzer to comply with the requirements shall be cause for rejection of that lot of howitzers.

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4.5.3.3 Fire control testing. Each completely assembled howitzer shall be tested for fire control (see 3.3.2.1 through 3.3.2.6) using the methods of inspection specified in 4.6.2. Failure of the fire control to comply with the requirements shall be cause for rejection.

4.5.3.4 Proof-firing testing. Unless otherwise specified (see 6.2), one (1) howitzer shall be randomly selected from each howitzer lot and subjected to proof firing in accordance with 3.3.3 using the test method specified in 4.6.3. Failure of the tested howitzers to comply with the specified requirements shall be cause for rejection of the lot. After correction of problem areas specified in 4.5.3.1, the resubmitted lot shall be tested as specified herein (paragraph 4.5.3.4) except that the sample size shall be three (3).

4.5.3.5 Road testing. Each howitzer which is proof fired in accordance with 4.5.3.4 shall also be subjected to a road test and brake system test to determine compliance with 3.3.4 using the test method specified in 4.6.4. Failure of the howitzer to comply with the requirements shall be cause for rejection of the howitzer lot.

4.5.3.6 Boresight retention testing. After proof-firing and road testing, the howitzer shall be tested for boresight retention to determine compliance with 3.3.5 using the test method specified in 4.6.5. Failure to maintain boresight setting after proof-firing and road testing shall be cause for rejection of the howitzer lot.

4.5.4 Inspection equipment. Unless otherwise specified herein, responsibility for acquisition, calibration, maintenance and disposition of acceptance inspection and test equipment prescribed on List of Inspection Equipment Numbers EL12008000-I and for all other inspection equipment required by applicable specifications, shall be in accordance with MIL-I-45607.

‘4.5.5 Inspection of packaging. Unless otherwise specified (see 6.2), inspection to determine compliance with preservation, packing and marking requirements of the applicable packaging documentation, for the level designated in the contract, shall be as specified in MIL-P-14232.

4.6 Methods of inspection.

4.6.1 Functioning test. The howitzer shall be tested as specified in 4.6.1.1 through 4.6.1.8.

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4.6.1.1 Elevating mechanism test. A torque wrench graduated in inch-pounds shall be utilized for measuring loads at the shaft of the left handwheel. Measurements shall be taken at the following elevations, both in elevating and depressing: -4.2° (-75 mils), +5° (89 mils), +10° (178 mils), +15° (267 mils), +25° (444 mils), +35° (622 mils), +45° (800 mils), +55° (978 mils), +65° (1156 mils) and +70° (1245 mils). The method for taking measurements is as follows: With the cannon and recoil mechanism installed and the carriage in emplaced position and level, depress to -75 mils and take the torque reading on the nut clockwise or counterclockwise. Then elevate to the next station and repeat taking torque readings on the nut clockwise or counterclockwise. Repeat this operation at all stations while elevating the cannon and recoil mechanism between -75 mils and maximum elevation +1245 mils.

4.6.1.1.1 Elevating mechanism backlash measurement test. The following procedure shall be utilized to measure left and right system backlash:

a. Equipment utilized:

1) A circular template, approximately 15 inches in diameter, with clearance cut-outs for the handle, located 180 degrees from each other, with degree markings on the front face, showing a span of 50 degrees and graduated in one degree intervals, located left and right of the zero mark and two mounting clamps on the rear face.

2) A pointer with an attached spring and chain.

3). Adial indicator (with a range per revolution of .100 inch and a total range of 1.00 inch)

4) A strap clamp with a wuick release tension hook

5) A split clamp with a quick release tension hook and with a contract surface for the dial indictor point.

b. Assembly of the test equipment to the howitzer to be inspected:

elevating mechanism handwheel by 1locating the shoulder, on thei back of the template, into the nut counterbore and securing it to the handwheel with both mounting clamps on the rear face of the template

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2) The pointer, for the circular template, with an attached spring and chain, is mounted behind the same handwheel and secured to the handwheel housing by pulling the chain firmly around the handwheel housing and hooking the chain in the notch provided as part of the pointer assembly.

3) The dial indicator is mounted to the ~~strap clamp~~ and then mounted on the lower end of the left cylinder (ref. dwg, 12007737) of the left elevating screw assembly (ref. dwg. 12007700).

4) The split clamp, with a quick release tension hook. is clamped in place on the lower end of the left tube (ref; dwg. 12007745) of the same elevating screw assembly just below the dial indicator with the contact surface of the clamp aligned with and against the dial indicator point. Note: This clamp is relocated on the tube after each elevation or depression setting of the cannon tube

c. Backlash Inspection procedure:

1) Set M17 quadrant to the desired elevation setting.

2) Disengage the clutch by pressing down on the clutch handle. Elevate or depress the weapon by turning the elevating handwheel, on either side of the carriage.

3) Turn the handwheel counterclockwise until the pointer is aligned with the zero degree marking on the circular template. While holding the handwheel steady at zero degrees, set the dial indicator to zero.

4) While observing the dial indicator, continue, turning the handwheel counterclockwise and stop when the dial indicator pointer has made at least one and a third revolutions around the dial past the original zero setting (approximately .130 inch).

5) continue watching the dial indicator and turn the handwheel in a clockwise direction and stop when the dial indicator pointer returns the one-and-a-third revolutions back to the original zero reading. Read and record the number of degrees of backlash on the circular template aligned with the pointer.

6) Read and record backlash readings at the following elevations and depressions; -75 mils, + mils, +267 mils, +622 mils, +978 mils and +1245 mils.

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7) Repeat the above procedure for the right side, reassembling the test equipment on the right side, for each elevation and depression as specified in step 7, in mils, taking care to move the split clamp, when required, to preclude damage to the dial indicator.

4.6.1.2 Traversing mechanism test. A torque wrench graduated in inch-pounds shall be utilized for measureing loads at the shaft of the handwheel. With howitzer in emplaced position and carriage level, traversing measurements shall be taken at 0°, + 15° and + 22° while traversing the howitzer with the tube—at zero degrees elevation. The handwheel shall not exceed the specified backlash.

4.6.1.2.1 Traversing mechanism backlash measurement test.

I. The following procedure shall be utilized to measure left and right system backlash.

a. Equipment utilized:

1) A circular (0.D. same diameter as traverse handwheel) template with clearance cut-outs for the handle, located 180 degrees from each other, with degree markings on the front face, showing a span of 50 degrees and graduated in one degree intervals located left and right of the zero mark and two mounting clamps on the rear face.

2) A pointer with an attached spring and chain.

3) A dial indicator (with a range per revolution of .100 inch and a travel range of 1.00 inch) attached to a magnetic base with an adjustable arm extension.

4) The large circular template utilized for the elevation backlash test.

b. Assembly of the equipment to the Howitzer to be tested:

1) The circular template is centrally mounted over the traverse handwheel and secured with the clamp provided.

2) The pointer is mounted behind the traverse handwheel and secured in place by pulling the chain firmly around the handwheel housing and hooking the chain to the notch provided.

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3) The large circular template that is utilized for the elevation backlash test is mounted on the left hand elevation handwheel.

4) The dial indicator, mounted on the magnetic base with adjustable arm/extension, is placed on the left hand top surface of the Bottom Carriage, adjacent to the traverse stop, with the indicator point contacting the large circular template on the elevation handwheel at right angles.

c. Backlash inspection procedure:

1.) With the howitzer in firing position, gun tube level, and the "V" notch on the rear of the top carriage base aligned at 400 roils on the azimuth plate located at the rear of the bottom carriage, turn the traverse handwheel clockwise to align the *zero* degree mark on the circular template with the pointer.

2) Holding the handwheel steady, position the magnetic base to depress the dial indicator point (use at least 3 revolutions of travel) and orient the indicator point at right angles to the circular template *on* the elevation handwheel. Set the indicator dial at zero.

3) While watchinq the dial, turn the handwheel clockwise at least one-and-a-third turns (on the dial).

4) While watching the dial indicator, turn the handwheel in a counterclockwise direction and stop when the dial indicator pointer returns the one-and-a-third revolutions (on the dial) hack to the original zero reading. Read the number of degrees of backlash on the circular template aligned with the pointer.

5) Record the reading on a suitable inspection form.

6) Ilemove the magnetic base and dial indicator from the bottom carriage.

7) Repeat steps one through six at 670 and 780 mils on the azimuth plate.

8) Traverse the top carriage to 20 mils on the azimuth scale.

9) Repeat steps one through six at 20 mils and 1.30 roils on the azimuth scale.

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4.6.1.3 Speed shift assembly test. The speed shift assembly is actuated by moving the hydraulic selector valve to "DOWN" position. With howitzer in enplaced position, elevate or depress the cannon tube to approximately 800 mils elevation to provide a balance for speed shifting the howitzer. Pump one or both hydraulic pumps to lower the speed shift, permitting the howitzer to be raised off the ground. Lift the right and left trails and pivot the howitzer 360°. Move the selector value to "UP" position. A controlled rate of descent to the firing base of not less than 5 seconds and not more than 20 seconds shall be maintained during the gravity drop phase. Pump the hydraulic pumps to retract the speed shift to allow the howitzer to be placed on the ground.

4.6.1.4 Variable recoil mechanism Test. Check the installation and adjustment of the variable recoil linkage. Recuperator cylinder marks at 1.05° and 55° shall remain visible within the notch on the regulator lever during repeated cvcling between minimum and maximum cannon tube elevations. The variable recoil mechanism shall move freely without binding or excessive looseness throughout the entire range of travel.

4.5.1.5 Wheel actuator test. The completed howitzer shall be assembled with the cannon tube and breech in the towel position (travel lock engaqed) and the parking brakes disengaged. Levers for the speed shift and wheels shall be in the "OFF" position. Turn lever to the "WHEELS DOWN" AND pump to release weight from the lock pins. Release the lock pins. Turn lever to "WHEELS UP" position. Carriage shall settle to baseplate in not less than 5 seconds and not more than 20 seconds. Pump until the wheels go up to the firing position and engage lock pins. Turn whwwl lever to "OFF"; turn the lever to "WHEELS UP". Pump to release weight from the lock pins." Release the lock pins. Turn the lever to "WHEELS DOWN" and pump until the wheels are in travel piosition and engage the lock pins. The maximum force applied on the hydraulic pump handle to raise and lower the carriage shall not exceed 32 pounds.

4.6.7.6 Locking device test. All locking and fastening devices shall be tested for secure assembly and positive action

4.6.1.7 Trunnin adapter Plates test. A calibrated gunner's quadrant and trunnion adapter gate inspection equipment (see 4.5.41) shall be used to determine trunnion cross-level and trunnion on adapter plate to cannon tube parallelism.

A 4.6.1.8 Handbrakes test. The handbrakes of the completed howitzer, shall be tested to determine compliance with 3.3.1.81 The requirement may also be tested by applying 2650 + 50 pounds push or pull, on the lunette with the howitzer in traveling position on a lrbrel surface.

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4.6.2 Fire control test.

4.6.2.1 Fire control test equipment and orientation of fire control instruments.

a. Mechanical jacks with a minimum of one square foot base for leveling the carriage trunnions.

b. A plumb line approximately .062 inch in diameter and about 45 feet long shall be suspended one-foot or less from the end of the muzzle brake. The plumb line shall be dampened.

c. A boresight target shall be constructed on non-warping and nonshrinking material and placed approximately 50 meters from the axis of trunnions. The test target pattern (Figure 1) shall be used for fire control and cannon bore alignment. The target shall be placed at the specified distance. The vertical line on the target shall be set plumb with the plumb line established for cannon tube plumb travel- The horizontal target line shall be established by the zero elevation setting of the cannon tube which will establish the required height of the target from the ground.

d. The borescope"used for cannon tube elevation plumb travel shall have a cross hair reticle mounted in a self-aligning adapter to be inserted into the breech of the cannon so that the reticle has a focusing range of five meters to infinity. The reticle of the telescope shall be collimated with respect to the outside diameter of the adapter in its installed position within + .05 mils. The borescope shall have magnification of four power or higher.

e. Peep sights having an aperture of 1/16 inch may be used to eliminate parallax in the panoramic telescope and elbow telescope, as necessary.

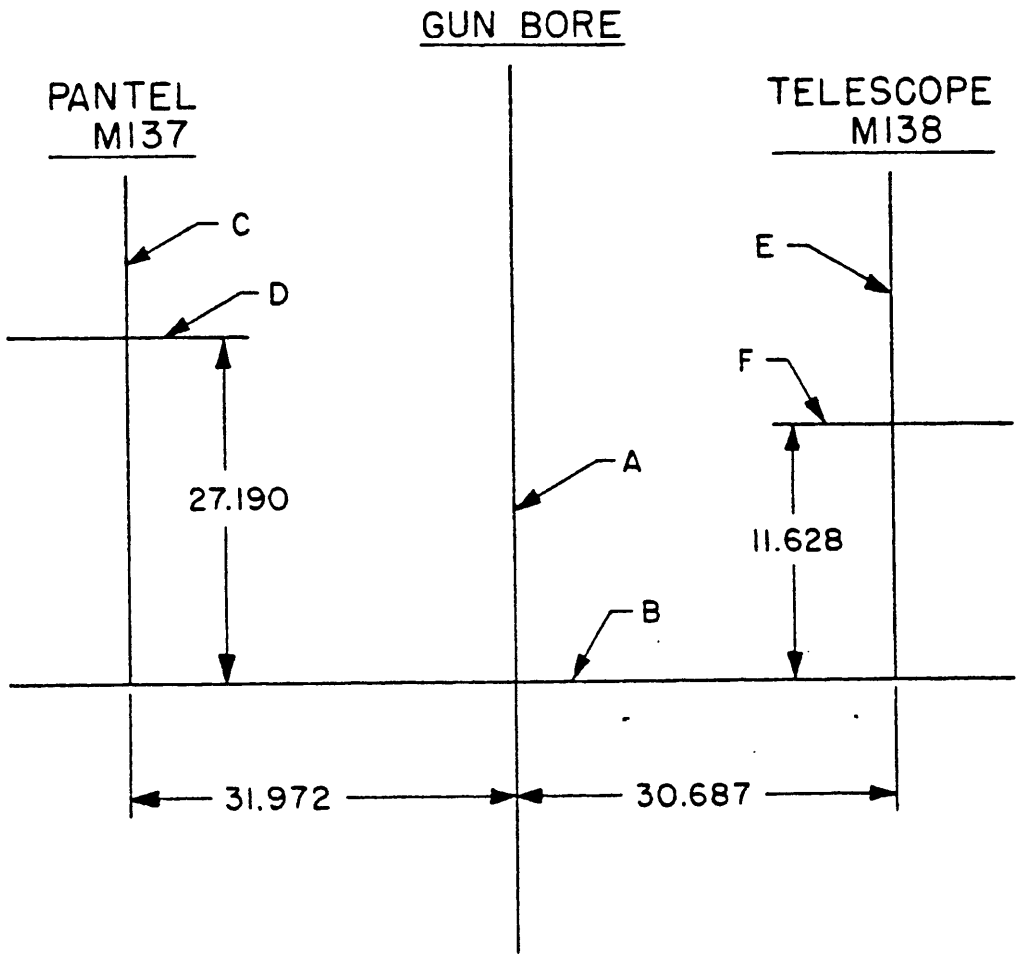
f. The carriage trunnions shall be cross-leveed as a preparatory step for cannon tube plumb travel.

g. Use cannon tube top leveling pad for zeroing out cannon tube. Only a calibrated gunner's quadrant shall be used as an inspection aid on the cannon tube pad.

h. Sight through the borescope and align the target lines A and B (Figure 2 in coincidence with the borescope reticle lines.

i. Set the level vials on the M171 and quadrant M17 to the level position (bubbles centered within the thickness (width) of a graduation line).

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Notes:

1. A & B Centerline of Gun Bore/Borescope
2. C & D Optical Axis of Pantel M137
3. E & F Optical Axis of Elbow Telescope M138

FIGURE 2

Target for fire control alignment to gun bore

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Sight through the panoramic telescope and align the reticle to the intersecting target lines C and D (Figure 1) with the elevation and azimuth knobs. Ascertain that the level vial bubbles for pitch and cross-level bubbles for pitch and cross-level are centered within the thickness (width) of a graduation line and the azimuth counter reads 3200 mils.

k. Set the reticle of the telescope M1.38 to the zero position and align to the target intersecting lines E and F (Figure l). For reticle/target line coincidence, make all adjustments with the boresight screws on the M172 mount.

l. The quadrant M18 level vial bubble shall be centered to thickness (width) of a graduation line with the elevation reading at 0000 mils. The cross-level vial bubble shall be centered to thickness (width) of a graduation line when boresight adjustments are set.

m. Counter backlash may be eliminated by always making the last movement in the elevation counter mechanism from high to low.

n. Backlash in the panoramic telescope M1.37 may be eliminated by always approaching the target from left to right.

4.6.2.2 Gun elevation-plumb travel. Each completely assembled howitzer shall be tested through the elevation range specified in 3.3.2.2. The cannon tube shall track a plumb line throughout the elevation range and deflection limit specified in 3.3.2.1. Any howitzer which does not track plumb within the specified limit shall be cause for rejection.

4.6.2.3 Alignment of mount M171., panoramic telescope M137 and quadrant M17.

a. The cannon tube shall be set at zero elevation and all level vial bubbles shall be centered within the thickness {width) of a graduation line.

b. The quadrant elevation level vial shall be adjusted as necessarv and shall meet the requirements specified in 3.3.2.2.a.

c. Subsequent to boresighting, the panoramic telescope shall align on the M139 alignment device at 4800mils within the limits specified in 3.3.2.2.b. Remove the alignment device before elevating the gun.

d. The gun tube shall be elevated from zero to 1275 mils in 200 mil increments. The pitch and cross-level knobs shall be rotated to relevel the mount and the level vial bubbles centered

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within the thickness (width) of a graduation line. Caution should be taken when rotating the knobs in one direction so as not to introduce backlash. If override shall occur, repeat the procedure with the applicable knob. At each 200 mil increment the level vials shall be centered within the thickness (width) of a graduation line and vertical line of the telescope[s] reticle shall not deviate more than the limits specified in s.3.2.2.c throughout the range of zero to 899 mils. Continue the above procedure starting at 900 roils to 1.275 mils in 200 mil increments. For the higher elevations, deviation shall not exceed the limits specified in 3.3.2.2.c.

4.6.2.4 Alignment of telecope M138, mount M1.72 and quadrant M1.8 .

a. The cannon tube shall be set at zero elevation. place a calibrated gunner’s quadrant on the pads provided on the mount. The elevation pads shall be parallel to the cannon tube within the limits specified in 3.3.2.3.a.

b. Set the quadrant to read 0000. The elevation level vial bubble shall be centered within the limits specifid in 3.3.2.3.b, as measured with the calibrated gunner's quadrant.

4.6.2.5 Quadrants M17/M18 elevation counter accuracy. Prior to proof firing, the elevation counter's accuracy shall not exceed the deviation limit specified in 3.3.2.4, excluding backlash at any cannon tube elevation setting. Subsequent to proof firing, the deviation shall not exceed maximum limit specified in 3.3.2.4, excluding backlash.

Backlash-quadrants M17/M18. The elevation knobs on each quadrant shall be tested for maximum allowable backlash at any elevation setting. Backbash;) clockwise and counterclockwise directions. Backlash measurements shall be taken using a calibrated gunner's quadrant. Backlash shall not exceed the limit specified in 3.3.2.5.

4.6.2.7 Operability and illumination. Mounted fire control shall have a visual and tactile examination for knob function to detect looseness, binding or excessive torque. Counters shall operate properly and the numerals shall be legible and readable at light levels ranging from dusk into darkness. Level vial bubbles and graduation lines shall be clear and discernible at light levels ranging from dusk into darkness. Radioactive illuminated retitles in the panoramic and elbow telescopes shall have clear and *concise* line patterns that are discernible at light levels ranging from dusk into darkness. Off carriage fire control shall be examined in the same manner as indicated above.

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4.6.3 Proof firing test. Each howitzer shall be proof fired as specified in Table I. Data shall be recorded on all rounds for breech pressure, recoil cycle time, rod pull curves~ recoil length and terminal velocity in counterrecoil. Rounds shall be fired in the specified sequence, however, if the test will unreasonably be prolonged due to range restrictions, the sequence for round numbers three (3) through eight (8) may be rearranged . Additionally, if there is a delay between rounds greater than 10 minutes, one M119A1 warmer must be fired before the next test round is fired. Rounds, number 1 and 2, are warmers and acceptance shall not be based on results from these unless structural or functional failure occurs. At the completion of the test, a comprehensive visual-manual examination shall be conducted to determine compliance with the requirements. If any of the requirements of paragraphs 3.3.3 through 3.3.3.5 are not met, the howitzer shall be rejected.

TABLE I. Proof firing

Round Number	Prop	Zone	Proj. Inert	QE and variable Recoil Setting	Traverse
1	M4A2	7	2/	0-90 Mils/Long Kecol	
2	M119A1	8	2/	0-90 Ails/Long Recoil	
3	*M 2 0 3 /*	8S	2/	0-90 Mils/Long Recoil	Max. Right
4	M203&/	8S	2/	900 Mils/Short Recoil	Max. Right
5	M203~/	8S	2/	12C0 Mils/Short Recoil	Max. Right
6	M203~/	8S	2/	0-90 Mils/Long Recoil	Max. Left
7	M203~/	8S	2/	900 Mils/Short Recoil	Max. Left
8	M203~/	8s	2/	1200 Mils/Short Recoil	Max. Left

1/ Charge shall be conditioned and fired at +145°F.

2/ M483A1 Projectile, w/fuze loaded to 104 + .5 pounds or equivalent.

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4.6.4 Road test. After proof firing, the howitzer, in traveling position with telescopes in stowed position,)ess dummy round) shall be coupled to a vehicle equipped with standard pintle and road tested at a speed of not less than five mph over a hard, smooth surface. Application of the brakes shall cause the wheels to lock and slide. The break away emergency action of the brakes shall be tested by disconnecting the air hose couplings with the drain cock closed. The brakes shall apply immediately, causing the wheels to slide. Visual examination for cracks, weld failures, deformations and other forms of damage shall be conducted.

4.6.5 Boresight retention test. All lines of sight of the optical instruments includlng borescope, shall be secured in coincidence with their respective aiming points as defined in 4.6.2.1. The exact position of the carriage shall be marked to insure repositioning-after performing the road test and proof-firing test to determine possible displacement of any lines of sight. Upon completion of the proof-firing test (see 4.6.4), the carriage shall be positioned over the original marked location and the reticle of the borescope shall be placed in coincidence with its respective target aiming point (lines A and B of Figure 1). Observe the respective lines of sight determine their relationship to the original settings. The panoramic telescope and direct fire telescope lines of sight shall not have deviated more than the tolerance specified in 3.3.5.

4.7 Reliability, maintainability and durability demonstration.
The howitzer demonstration will be conducted in accordance with the operational mode summary and reliability failure definition in the M198 scoring criteria. The test will be conducted by the Government for informational purposes and shall provide the equivalent of a 2,500 EFC round reliability test and a 3,000 EFC round durability test. The two tester may be combined Maintenance data will be collected according to the scoring criteria, and the MTTR will be calculated according to the scoring criteria. Howitzers exhibiting a reliability, maintainability or durability failure shall be cause for the contractor to perform a failure analysis in accordance with 4.5.3.1. The government represtative shall verify the results of the failure analysis and assure deficiencies causing the failure have been corrected on all weapons and components in process at the contractor’s plant.

4.8 Howitzer components. Examination and tests of the M45 recoil mechanism, M39 carriage, M199 cannon, M139 alignment device, M137 panoramic telescope, M17 quadrant, M18 quadrant, M171 mount, M138 elbow telescope and M172 mount shall be as specified in the applicable specification.

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5. PACKAGING

5.1 Pilot pack. A pilot pack shall consist of a complete howitzer preserved in accordance with Packaging Data Sheet P12008000 for the level of protection specified in the contract (see 6.2), packed level B and forwarded as specified in 3.1.

5.2 Levels A and B. Preservation, packing and marking shall be in accordance with Packaging Data Sheet P1.2008000 for the level of protection specified in the contract (see 5.2).

5.3 Repair parts. Repair parts shall be preserved, packed and marked in accordance with the Packaging Data Sheet or other Packaging requirements, as specified in the contract (see 6.2).

6. NOTES

6.1. Intended use. The M198 Howitzer is a lightweight, towed, 155MM field artillery weapon suited for close support operations. This weapon is highly mobile, and is capable of traversing ± 22.5 degrees on-carrage and 360 degrees off-carriage with speed shift system, with an elevation range of minus 75 roils to plus 1.275 roils. This weapon is also capable of being air-lifted and transportable by helicopter.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. List of drawings, specifications and publications pertinent to the howitzer, showing applicable revision dates.

c. Shipping instructions: for first article and pilot pack (see 3.7, 4.4 and 5.1).

d. Marking, if different, and block of serial numbers (see 3.6).

e. Number of howitzers required for proof-firing, if other than as specified (see 4.5.3.4),

f. Packaging *inspection,* if different (see 4.5.5).

g. Selection of applicable level of protection, A or B (see 5.1 and 5.2).

h. Packaging requirements for repair parts (see 5.3)s

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6.3 Firing records. One (1) copy of each final Proving Ground Firing Record for each howitzer tested shall be sent to: AMCCOM, ATTN: AMSMC-QAR-Q and AMSMC-LW-A, Dover, New Jersey for informational purposes.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.5 Supersession data. This specification supersedes MIL-H-45984A (WC), dated 15 February 1977, with Amendment 7, dated 03 November 1982: Howitzer, Medium, Towed, 155MM: M198:

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1025-A142)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-45984B	HOWITZER, MEDIUM, TOWED, 155MM, M198

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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